PEARSON EDEXCEL INTERNATIONAL GCSE

# Mathematics A

## Unit 1 • Chapter OL-T20 Classified

**Cross-session • chapter-organised candidate**

7 classified items • 24 marks • 1 chapter(s)

**Source-faithful practice with synchronized solutions**

**Every question is followed by its worked answer/reference**


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**ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE**

# Contents

Unit 1 • Unit 1 • Chapter OL-T20 • 7 items • 24 marks



June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

# Completing the Square

**Unit 1 • 24 marks • 7 item(s)**

**June 2025 | 4MA1-2H Q25 | 3 marks**
Negative leading coefficient or maximum form

**June 2024 | 4MA1-1H Q25 | 4 marks**
Negative leading coefficient or maximum form

**June 2023 | 4MA1-2H Q19 | 3 marks**
Positive non-unit leading coefficient

**June 2022 | 4MA1-1H Q24 | 4 marks**
Negative leading coefficient or maximum form

**January 2022 | 4MA1-1H Q20 | 4 marks**
Negative leading coefficient or maximum form

**November 2021 | 4MA1-1H Q17 | 3 marks**
Positive non-unit leading coefficient

**November 2025 | 1H Q21 | 3 marks**
Monic quadratic x^2+bx+c

**Original question context and synchronized companion pages follow.**

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

## Working Unit 27

4MA1-2H Q25

ORIGINAL QUESTION
**3 marks**

**25** (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where $a$, $b$ and $c$ are integers.

.......................................................

**(3)**

## Worked Solution 27

Complete the square

UNIT 1
**3 marks**

### Rearrange

$$28 + 24x - 6x^2 = 28 - 6(x^2 - 4x).$$

### Complete

$$28 - 6\big((x-2)^2 - 4\big) = 52 - 6(x-2)^2.$$

### Final answer

$$52 - 6(x-2)^2 \quad (a = 52,\ b = 6,\ c = 2)$$

## Question 16

4MA1-1H Q25

ORIGINAL QUESTION
**4 marks**

**25** $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where $a$, $b$ and $c$ are constants.

$f(x) =$ ..........................................................

## Worked Solution 16

Complete the square

MODULAR UNIT 1

**4 marks**

### Rearrange

$$17 - 3x^2 + 12x = 17 - 3[(x-2)^2 - 4].$$

### Final answer

$$29 - 3(x-2)^2$$

## Question 24

4MA1-2H Q19

ORIGINAL QUESTION
**3 marks**

**19** Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

..........................................................................

## Worked Solution 24

Complete the square

MODULAR UNIT 1
**3 marks**

### Group the quadratic terms

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5.$$

### Complete the square

$$3(x^2 - 2x) + 5 = 3[(x-1)^2 - 1] + 5 = 3(x-1)^2 + 2.$$

### Final answer

$$3(x-1)^2 + 2$$

## Original question 15

4MA1-1H Q24 Q24

ORIGINAL QUESTION
**4 marks**

**24** Express each of $a$, $b$ and $c$ in terms of $q$ so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ ......................................................

$b =$ ......................................................

$c =$ ......................................................

## Worked solution 15

Chapter: Completing the Square

MODULAR UNIT 1

4 marks

### Q24 – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

**Method**

q + 12x - qx^2 = -q[x^2 - (12/q)x] + q.

*Why: Factor -q from the quadratic and linear x-terms.*

**Method**

x^2 - (12/q)x = (x - 6/q)^2 - 36/q^2.

*Why: Complete the square by halving the coefficient of x.*

**Method**

Therefore q + 12x - qx^2 = -q(x - 6/q)^2 + 36/q + q.

*Why: Substitute the completed-square identity and distribute -q.*

**Method**

Matching a - b(x - c)^2 gives a = q + 36/q, b = q, and c = 6/q.

*Why: Compare the constant, outside coefficient, and horizontal-shift terms.*

**Final answer**

**a:** a = q + 36/q

**Final answer**

**b:** b = q

**Final answer**

**c:** c = 6/q

## Original question 16

4MA1-1H Q20 Q20(a), Q20(b)

ORIGINAL QUESTION
**4 marks**

**20** (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where $a$, $b$ and $c$ are integers.

..........................................................................

**(3)**

**C** is the curve with equation $y = 7 + 12x - 3x^2$
The point $A$ is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of $A$

(.................... , ....................)

**(1)**

## Worked solution 16

Chapter: Completing the Square

MODULAR UNIT 1
**4 marks**

### Q20(a) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

**Method**
$7+12x-3x^2=7-3(x^2-4x)$.
*Why: Factor −3 from the quadratic terms.*

**Method**
$x^2-4x=(x-2)$^2−4.
*Why: Complete the square.*

**Method**
Therefore $7+12x-3x^2=19-3(x-2)$^2.
*Why: Substitute the completed square and simplify.*

**Final answer**
**completed square:** 19−3(x−2)^2

### Q20(b) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

**Method**
The maximum occurs when (x−2)^2=0, giving A=(2,19).
*Why: The negative coefficient means the squared term is subtracted from 19.*

**Final answer**
**maximum point:** (2,19)

(c) Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$ where $a$, $b$ and $c$ are integers.

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**(3)**

## Worked solution

November 2021 | Question 17 | 3 marks

Family: Complete the square for a non-monic quadratic • Variant: Positive non-unit leading coefficient

Method / working
1. (c)
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. $4x^2-8x+7=4(x^2-2x)+7$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. $=4\big((x-1)^2-1\big)+7$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $=4(x-1)^2+3$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. Answer: $(6y+5)(y-1)$, $f=frac{8w-3}{w+2}$, $4(x-1)^2+3$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: $(6y+5)(y-1)$, $f=frac{8w-3}{w+2}$, $4(x-1)^2+3$

**November 2025 | 1H | Question 21 | 3 marks**

Chapter OL-T20: Completing the Square

**21** Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where $a$, $b$ and $c$ are integers.

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**(Total for Question 21 is 3 marks)**

## Worked solution

November 2025 | Question 21 | 3 marks

Family: Complete the square for a monic quadratic • Variant: Monic quadratic $x^2+bx+c$

Method

$5x^2-20x+23=5(x^2-4x)+23$

$=5((x-2)^2-4)+23$

$=5(x-2)^2-20+23$

$=5(x-2)^2+3$

Answer: $5(x-2)^2+3$

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION